

Skills Summary

Solving AAS and ASA Triangles with the Law of Sines

Use this reference while completing your worksheet or when studying.

Skill 1 — The third angle first

Angle sum: $A + B + C = 180^\circ$. Two angles always give the third.

Naming: side a lies opposite angle A , b opposite B , c opposite C . A side and the angle across from it are a *pair*.

Two angles fix the shape; one side fixes the size.

Example: A triangle has $A = 47^\circ$, $B = 68^\circ$ and $c = 12$. Find C .

Step 1. Two angles are known and the third is asked for, so this is the angle sum, not the law of sines — the side plays no part.

Step 2. $C = 180^\circ - 47^\circ - 68^\circ \Rightarrow C = 65^\circ$

Try it: A triangle has $A = 39^\circ$, $C = 84^\circ$ and $b = 9$. Find B .

check: $B = 57^\circ$

Skill 2 — AAS: the side already has a partner

Law of sines: $\frac{a}{\sin A} = \frac{b}{\sin B} = \frac{c}{\sin C}$.

AAS means the known side is opposite one of the known angles, so it already has a partner and you can cross straight to the side you want:

$$\text{wanted side} = \frac{\text{known side} \times \sin(\text{angle opposite the wanted side})}{\sin(\text{angle opposite the known side})}$$

Example: $A = 36^\circ$, $B = 71^\circ$, $a = 10$. Find b .

Step 1. The known side a sits opposite the known angle A , and the wanted side b sits opposite the known angle B — two complete pairs, so no third angle is needed.

Step 2. $b = \frac{10 \sin 71^\circ}{\sin 36^\circ} = \frac{9.4552}{0.5878} \Rightarrow b \approx 16.09$

Try it: $B = 41^\circ$, $C = 63^\circ$, $b = 15$. Find c .

check: $c \approx 20.37$

Skill 3 — ASA: find the missing angle first

ASA means the known side lies *between* the two known angles, so it has no partner yet. Compute the third angle with the angle sum, and the side gains one.

Never pair a side with an angle beside it — only with the angle across from it.

Example: $A = 48^\circ$, $B = 57^\circ$, $c = 20$. Find a .

Step 1. The known side is c , but C is not given, so the side has no partner — the angle sum has to come first.

Step 2. $C = 180^\circ - 48^\circ - 57^\circ = 75^\circ$, then $a = \frac{20 \sin 48^\circ}{\sin 75^\circ} \Rightarrow a \approx \mathbf{15.39}$

Try it: $B = 34^\circ$, $C = 76^\circ$, $a = 18$. Find b .

check: $b \approx 12.01$

Skill 4 — Surveying with a baseline

A surveyor measures a **baseline** between two stations and the angle from each station to a distant point. That is ASA: the baseline sits between the two measured angles.

Check the answer: the longest side must face the largest angle. If your computed distance breaks that, a pair was crossed wrongly.

Example: Stations A and B are $c = 400$ m apart. The angle at A is 29° and at B is 54° . Find the distance from B to the point C .

Step 1. The distance from B to C is the side opposite A , and the baseline lies between the two measured angles — ASA, so the third angle comes first.

Step 2. $C = 180^\circ - 29^\circ - 54^\circ = 97^\circ$, then $a = \frac{400 \sin 29^\circ}{\sin 97^\circ} \Rightarrow a \approx \mathbf{195.38}$ metres.

Try it: Stations A and B are $c = 250$ m apart, with $A = 41^\circ$ and $B = 36^\circ$. Find the distance from A to C .

check: $b \approx 150.81$ metres